Getting Insights from Data: Descriptive Statistics

Learning Outcome

By the end of the lesson, the student will be able to…

- Describe and differentiate the scale types
- Understand what is descriptive univariate analysis
- Understand descriptive bivariate analysis

Scale Type

Qualitative

- Categorizes data in nominal or ordinal way.
- Nominal - Cannot be reordered according how big or small.
  e.g.: Gender, hair color, nationalities, name of people, etc.
- Ordinal - Can be reordered.
  e.g.:: measures of satisfaction, happiness,

Quantitative

There are two types of scale for quantitative data

- Absolute (ratios)
- Relative (intervals)

The difference between them is that in absolute scales there is absolute zero while in relative scales there is no absolute zero

Absolute Zero

When the attribute “height” is **zero** it means there is **no height**. This is also true for the weight.

Absolute

But for the temperature, when we have **0°C** it **does not mean there is no temperature**.

Relative

Relation of Scales Type

Nominal → Ordinal → Relative → Absolute

- Nominal : Related to their similarity (equal to/ not equal to)
- Ordinal : Check their order (larger than/ equal to/ smaller than)
- Relative : can be added to, subtract from
- Absolute : can see how many times value is larger (multiply), smaller (divide) than the other

Descriptive Univariate Analysis

Univariate Frequencies

Univariate is a term commonly used in statistics to describe a type of data which consists of observations on only a single characteristic or attribute. **(Counter)**

1 Absolute Frequency

Counts how many times a value appear.

2 Relative Frequency

Counts the percentage of times that value appears.

3 Absolute cumulative

The number of occurrences less than or equal to a given value

4 Relative cumulative

The percentage of occurrences less than or equal to a given value

Dataset

Friend	Max temp (°C)	Weight (kg)	Height (cm)	Gender	Company
Andrew	25	77	175	M	Good
Bernhard	31	110	195	M	Good
Carolina	15	70	172	F	Bad
Dennis	20	85	180	M	Good
Eve	10	65	168	F	Bad
Fred	12	75	173	M	Good
Gwyneth	16	75	180	F	Bad
Hayden	26	63	165	F	Bad
Irene	15	55	158	F	Bad
James	21	66	163	M	Good
Kevin	30	95	190	M	Bad
Lea	13	72	172	F	Good
Marcus	8	83	185	F	Bad
Nigel	12	115	192	M	Good

Calculate the univariate absolute and relative frequencies for height

Height	Absolute Frequency	Relative Frequency	Absolute Cumulative Frequency	Relative Cumulative Frequency
158				
163				
165				
168				
172				
173				
175				
180				
185				
190				
192				
195				

Univariate absolute and relative frequencies for height

Height	Abs. freq.	Rel. freq.	Abs. cum. freq.	Rel. cum. freq.
158	1	$1/14 = 7.14\%$	1	7.14%
163	1	7.14%	2	14.28%
165	1	7.14%	3	21.42%
168	1	7.14%	4	28.56%
172	2	14.29%	6	42.85%
173	1	7.14%	7	49.99%
175	1	7.14%	8	57.13%
180	2	14.29%	10	71.42%
185	1	7.14%	11	78.56%
190	1	7.14%	12	85.70%
192	1	7.14%	13	92.84%
195	1	7.14%	14	99.98%

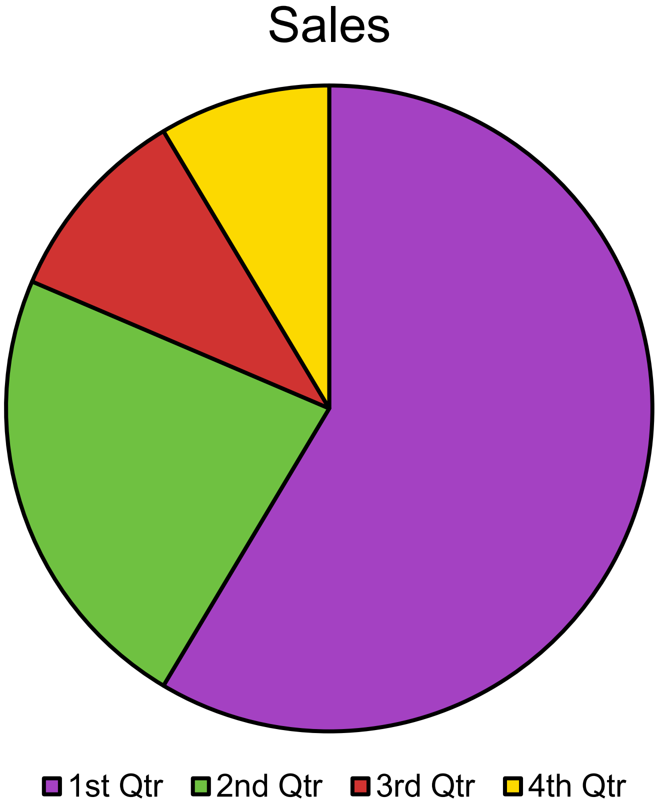
185	1	7.14%	11	78.56%
190	1	7.14%	12	85.70%
192	1	7.14%	13	92.84%
195	1	7.14%	14	$\boxed{99.98\%}$

The value of the relative cumulative frequency of the last row is **always 100%,** although perhaps with some decimal differences due to rounding of intermediate values

Univariate Data Visualisation

Pie charts

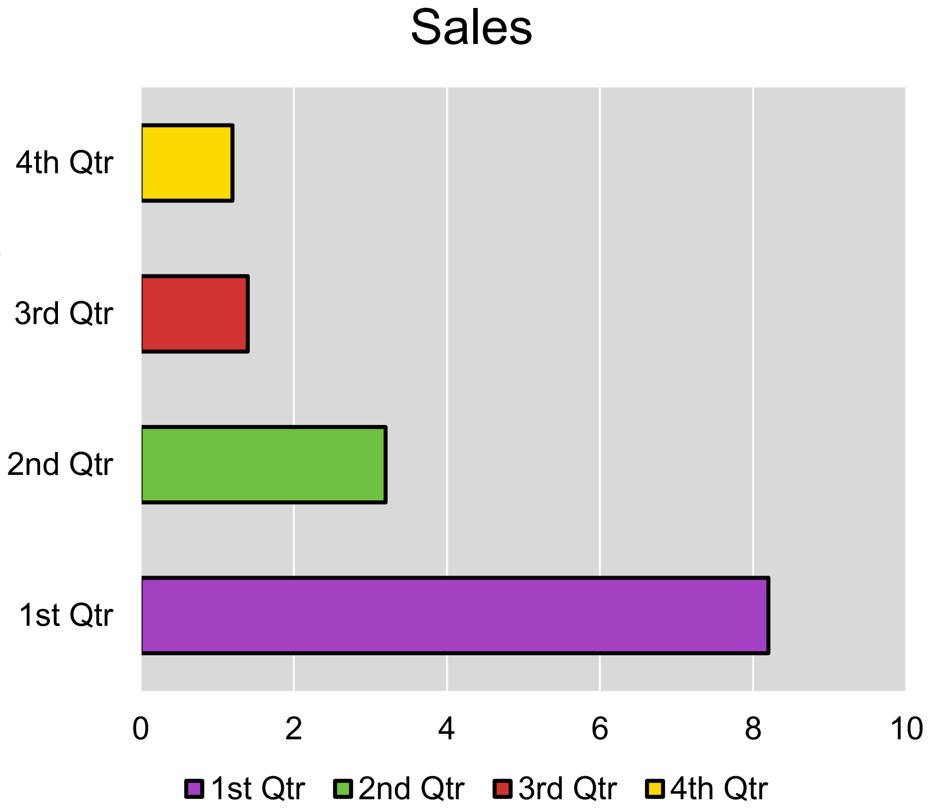
- For nominal scales
- Ordinal and quantitative scale possible (BUT NOT ADVISABLE)



Univariate Data Visualisation

Bar charts

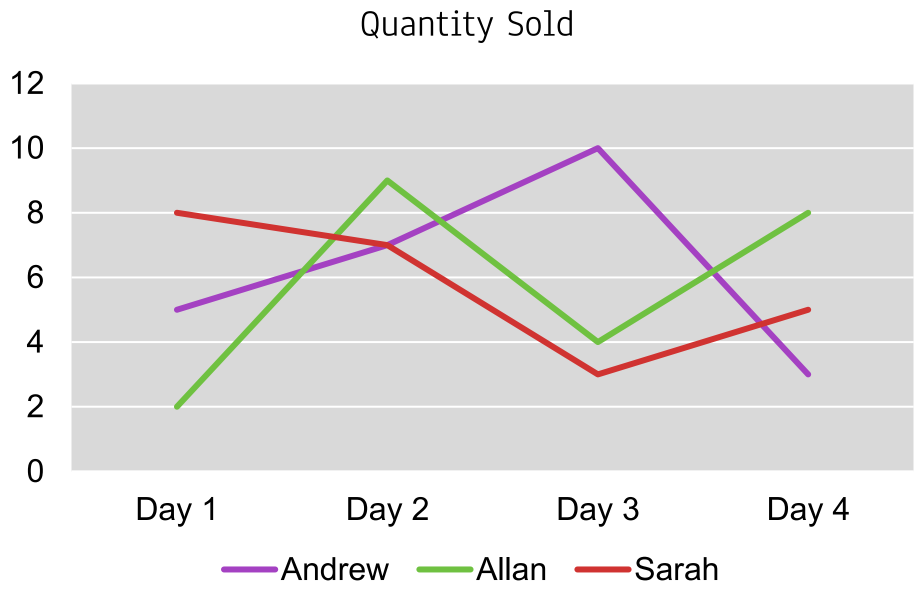
- For qualitative scales
- In horizontal bar, in increasing order



Univariate Data Visualisation

Line charts

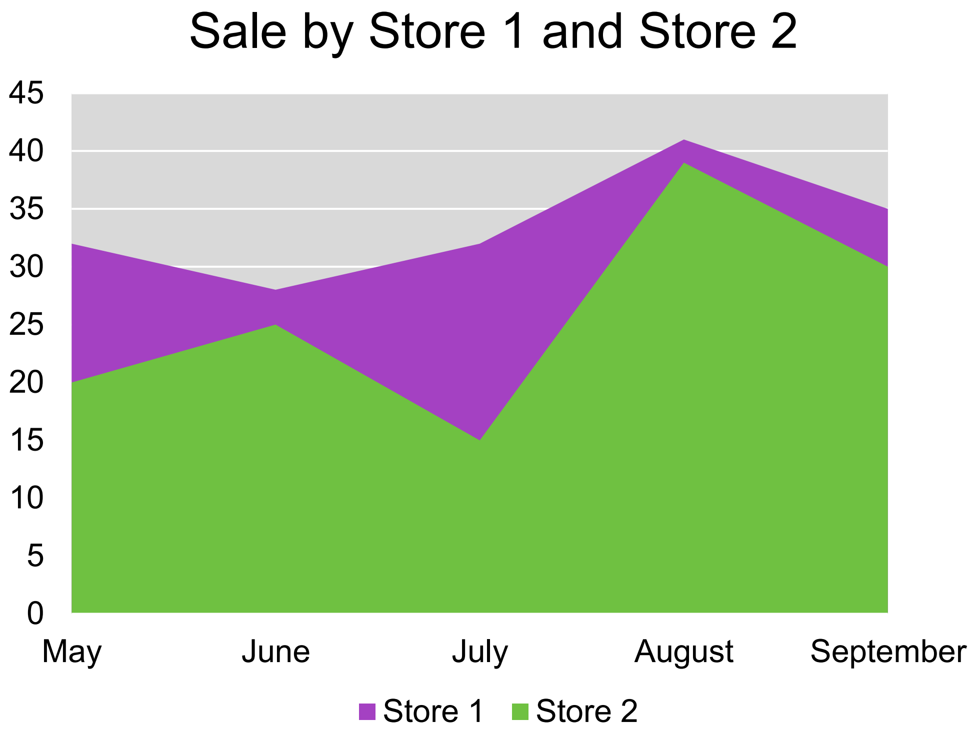
- For quantitative scale
- Deal with notion of time



Univariate Data Visualisation

Area charts

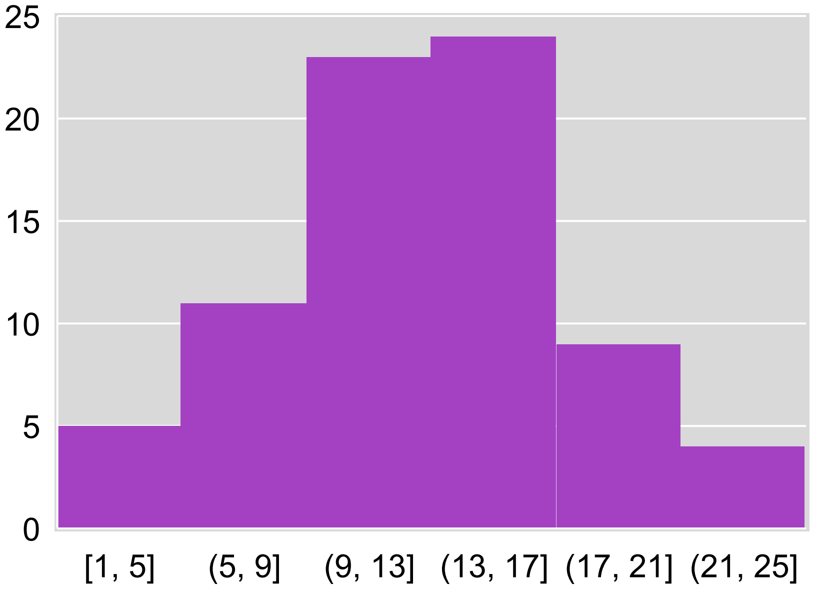
- Compare time series and distribution functions

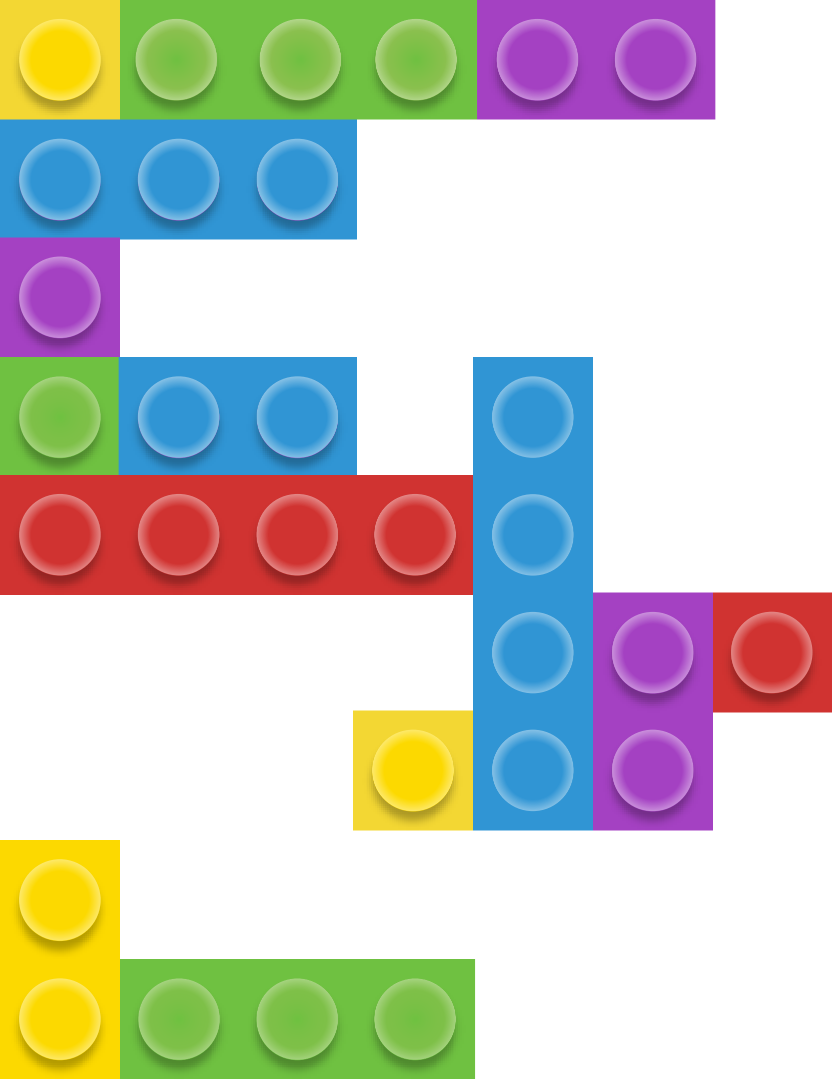


Univariate Data Visualisation

Histograms

- Represent empirical distribution for attributes with quantitative scale
- Characterized by grouping values in cells





Univariate Statistics

Statistics is a descriptor-it describes numerically a characteristic of the sample or the population.

Groups of univariate statistics:

- Location statistics
- Dispersion statistics

Location Univariate Statistics

- Identify a value in a certain position.

Minimum: the **lowest** value

Maximum: the **largest** value

Mean: the average value

Mode: the most frequent value

1st quartile: the value that is **larger than 25%** of all values

Median/ 2nd quartile: the value that is larger than **50%** of all values; the value that splits the sequence into two equal-sized sub-sequences

3rd quartile: the value that is larger than **75%** of all values.

Calculate the location univariate statistics for weight

Friend	Max temp (°C)	Weight (kg)	Height (cm)	Gender	Company
Andrew	25	77	175	M	Good
Bernhard	31	110	195	M	Good
Carolina	15	70	172	F	Bad
Dennis	20	85	180	M	Good
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Calculate the location univariate statistics for weight

Location statistics	Weight (kg)
Min	
Max	
Average	
Mode	
1st Quartile	
Median	
3rd Quartile	

1. Sort the values in ascending order.

$$55, 63, 65, 66, 70, 72, 75, 75, 77, 83, 85, 95, 110, 115$$

$$n = 14$$

2. First quartile, Q_1

$$Q_1 \Rightarrow \frac{1}{4}(n+1) = \frac{1}{4}(14+1) = 3.75$$

Q_1 is located between T_3 and T_4

$$3.75$$

$$3 + 0.75$$

T_3

65

$0.75(T_4 - T_3)$

$0.75(66 - 65)$

0.75

$$\therefore Q_1 = 65 + 0.75 = 65.75$$

3. Second quartile, Q_2 / Median

$$Q_2 \Rightarrow \frac{1}{2}(n+1) = \frac{1}{2}(14+1) = 7.5$$

Q_2 is located between T_7 and T_8

$$7.5$$

$$7 + 0.5$$

T_7

75

$0.5(T_8 - T_7)$

$0.5(75 - 75)$

0

$$\therefore Q_2 = 75 + 0 = 75$$

4. Third quartile, Q_3

$$Q_3 \Rightarrow \frac{3}{4}(n+1) = \frac{3}{4}(14+1) = 11.25$$

Q_3 is located between T_{11} and T_{12}

$$11.25$$

$$11 + 0.25$$

T_{11}

85

$0.25(T_{12} - T_{11})$

$0.25(95 - 85)$

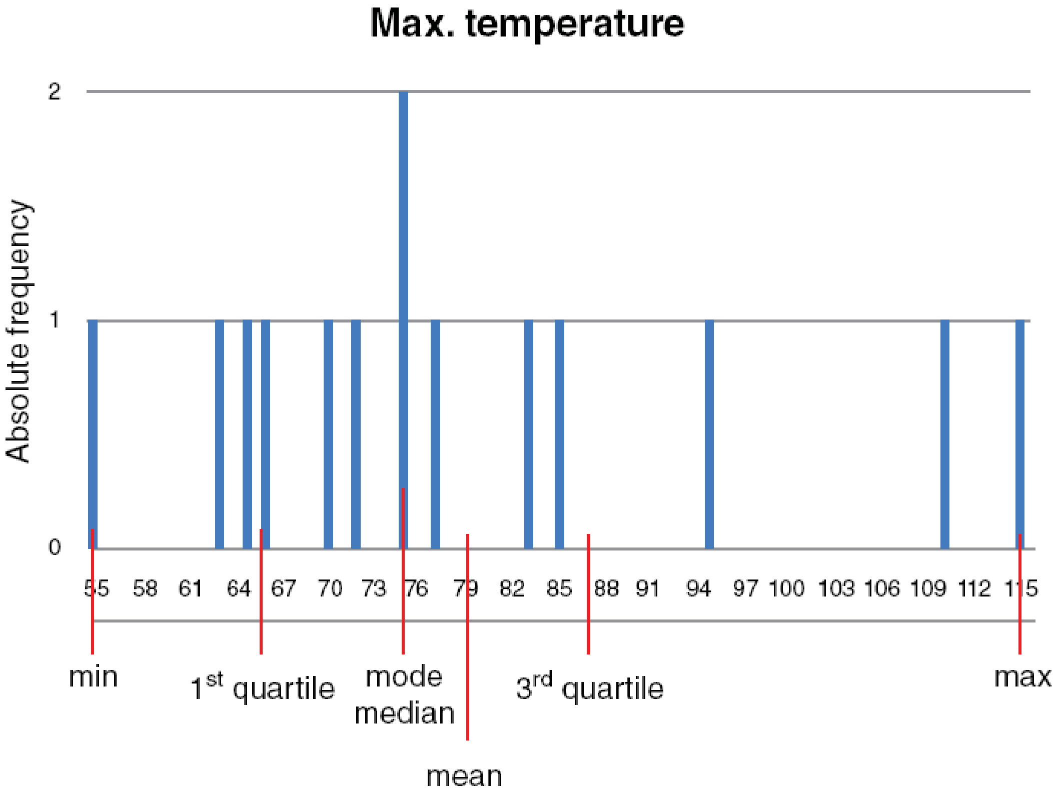
2.5

$$\therefore Q_3 = 85 + 2.5 = 87.5$$

Location univariate statistics for weight

Location statistic	Weight (kg)
Min	55.00
Max	115.00
Average	79.00
Mode	75.00
First quartile	65.75
Median or second quartile	75.00
Third quartile	87.50

Location statistics on the absolute frequency plot for attribute "weight"



Dispersion Univariate Statistics

- Measures how distant different values are.
- All of these dispersion statistics are only valid for quantitative scales

Amplitude : the **difference** between the **maximum** and the **minimum** values

Interquartile range (IQR): is the **difference** between the values of the **third** and **first quartiles**

Mean absolute deviation (MAD): a measure for the **mean absolute distance** between the **observations** and the **mean**

Standard deviation: another measure for the typical **distance** between the **observations** and their **mean**.

Amplitude : max-min

IQR: 3rd Q - 1st Q

Sample Mean absolute deviation (MAD)

$$\overline{MAD}_x = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n-1}$$

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Sample Standard Deviation

Variance

$$Variance = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$$

$n = number\ of\ observation$

$\bar{x} = mean\ value\ of\ the\ sample$

Calculate the dispersion univariate statistics for weight

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Dispersion Univariate Statistics

Location univariate statistics for weight

Location statistic	Weight (kg)
Min	55.00
Max	115.00
Average	79.00
Mode	75.00
First quartile	65.75
Median or second quartile	75.00
Third quartile	87.50

Statistics	Weight (kg)
Amplitude	60.00
IQR	21.75
$\overline{MAD}$	14.31
s	17.38

Mean Absolute Deviation, $\overline{MAD}$

$$\overline{MAD} = \frac{|55-79|+|63-79|+\ldots+|115-79|}{14-1}$$

$$= \frac{186}{13} = 14.31$$

Sample Standard Deviation, s

$$s = \sqrt{\frac{(55-79)^2 + (63-79)^2 + \ldots + (115-79)^2}{14-1}}$$

$$= \sqrt{\frac{3928}{13}} = 17.38$$

Common Univariate Probability Distributions

Uniform

- It is a very **simple distribution**
- The frequency of occurrence of the values is uniformly distributed in a given interval of values.
- An attribute x that follows a uniform distribution with parameters a and b, respectively the minimum and maximum values of the interval.

$$x \sim \mathcal{V}\,(a, b)$$

Normal

- Also known as **Gaussian distribution**, is the most common distribution, at least for continuous attributes.
- Physical quantities that are expected to be the sum of many independent factors.
- Example: people's heights or the perimeter of 30-year-old oak trees typically have approximately normal distributions.

Descriptive Bivariate Analysis

It is about pairs of attributes and their relative behavior

Descriptive Bivariate Analysis

- It is one of the **simplest forms** of statistical analysis,
- Used to find out if there is a **relationship between two sets of values**.
- Usually involves the variables X and Y.

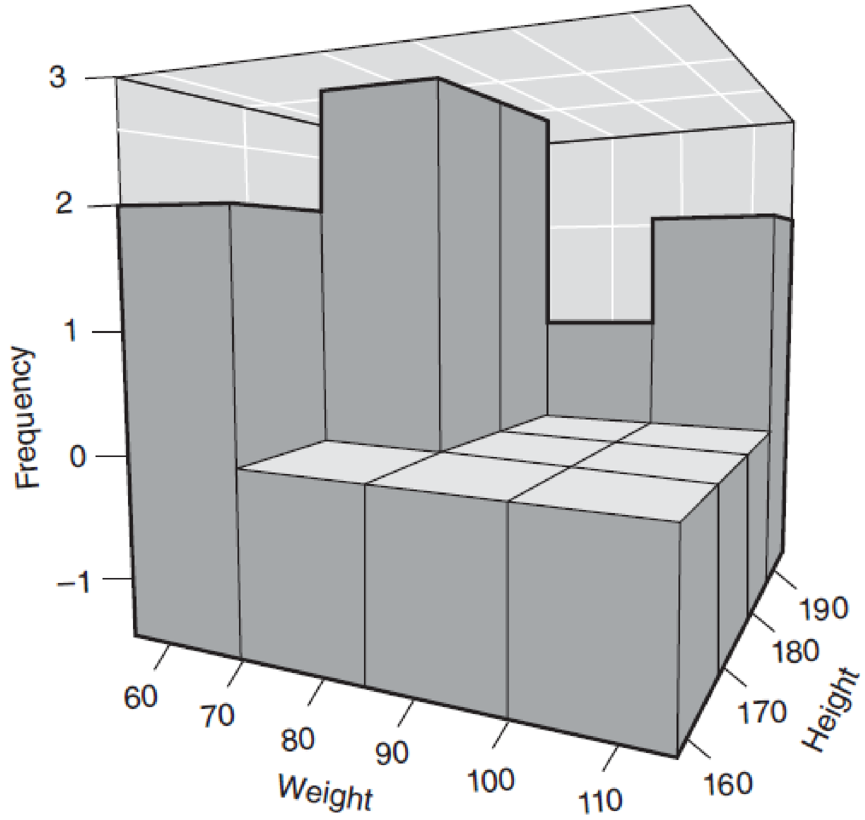
Calorie Intake, X	Weight (kg), Y
3500	110
2000	85
1500	64
2250	91
4500	130

- **Univariate analysis** is the analysis of one ("uni") variable.
- **Bivariate analysis** is the analysis of exactly two variables.

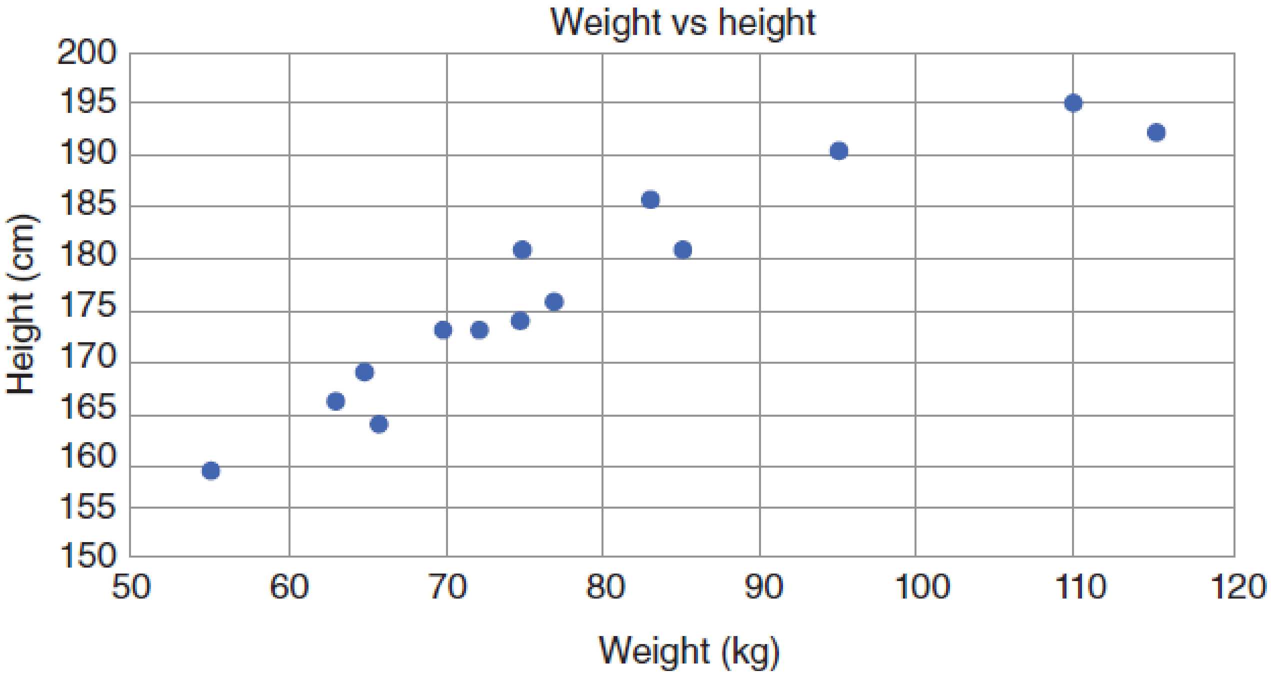
Two Quantitative Attributes

- In a data set whose objects have n attributes, each object can be represented in a n dimensional space:
  - A space with n axes, each axis representing one of the attributes .
  - The position occupied by an object is given by the value of its attributes

- Several **visualization techniques** that can visually show the distribution of points with two quantitative attributes.
  - Extension of the histogram called a **3D histogram**
  - **Scatter plots** illustrate **how the values of two attributes are correlated**. They make it possible to see how an attribute varies according to the variability of the other attribute.



3D histogram for attribute height and weight



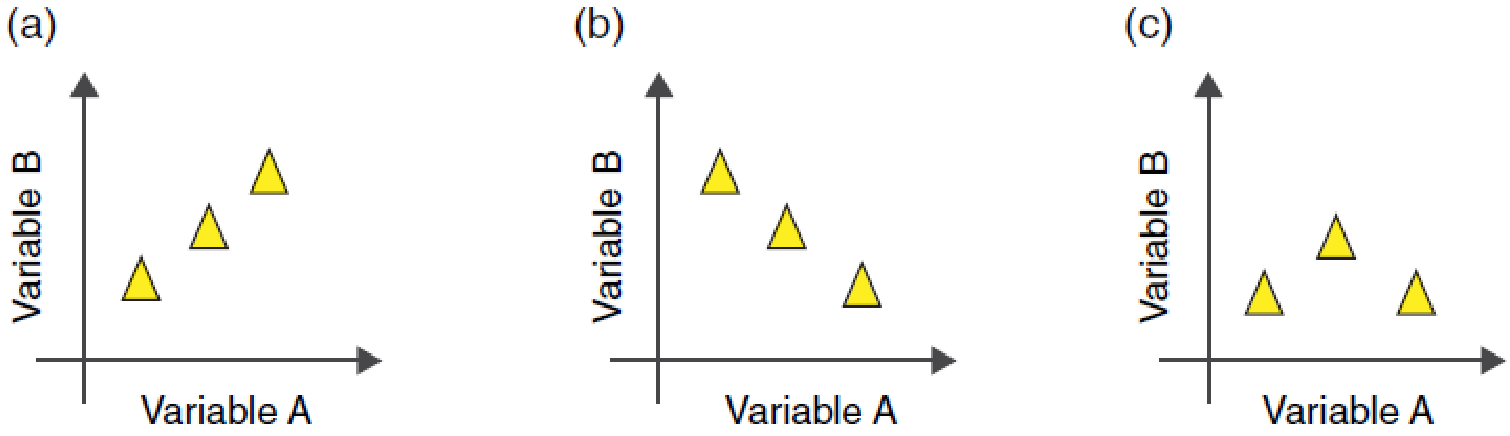
Scatter plot for attribute height and weight

It can be seen as a general tendency that people with larger weights are taller, and people with smaller weights are shorter

Two Quantitative Attributes

- Covariance
  - It is a measure of the relationship between **two random variable**
  - Useful measure to show **how values of two attributes relates to each other**
  - **Size of the range of values** of attributes influence the covariance value
- Correlation
  - **Pearson correlation:** linear correlation between two attributes
    1. Positive correlation if the value is 1
    2. Negative correlation if the value is -1
    3. No correlation if the value is 0

Pearson correlation



Positivie correlation

Nagative correlation

No correlation

Pearson correlation formula

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

r = correlation coefficient

x_i = values of the x-variable in a sample

$\bar{x}$ = mean of the values of the x-variable

y_i = values of the y-variable in a sample

$\bar{y}$ = mean of the values of the y-variable

Spearman's rank correlation

- It is based on ranking.
- It compares ordered lists of each of the two attributes
- Formula is similar used to calculate the Pearson correlation
- coefficient, but instead of using the values , it uses the order of the values in the rank (rs), rx and ry respectively
  - A rs of 1 indicates a perfect association of ranks,
  - A rs of zero indicates no association between ranks
  - A rs of -1 indicates a perfect negative association of ranks.
- The closer rs is to zero, the weaker the association between the ranks.

Spearman's rank correlation

1. Sort the height in ascending order
2. When there is a draw , the value used is the average of the positions they would have occupied.

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Two Qualitative Attributes
at least one of them Nominal

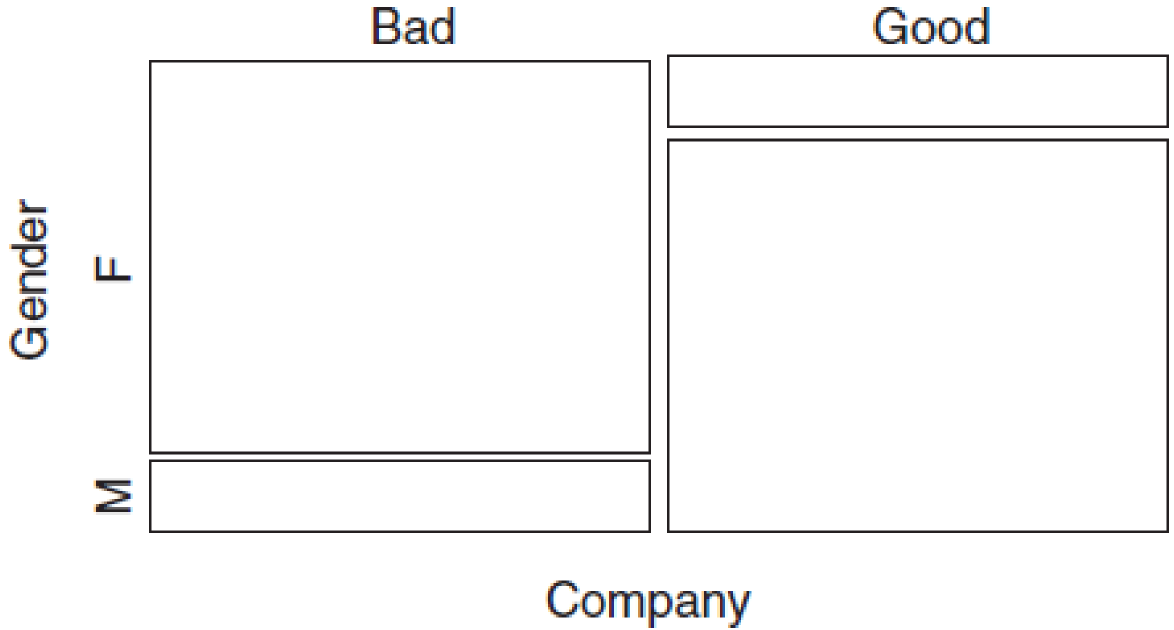
- When the attributes are both qualitative with at least one nominal, contingency tables are used.
- **Contingency tables -** present the joint frequencies, facilitating the identification of interactions between the two attributes.
- They have a **matrix-like** format, with cells in a square and labels at the left and top
  - Right most column indicates the totals per row
  - Bottom most row are the totals per column.
  - The bottom right-hand corner has the total number of values.

Contingency tables

		Company		
		Good	Bad	
Gender	Male	6	2 1	8 7
	Female	1	5 6	6 7
		7	7	14

Mosaic plots

Based on contingency tables, showing the same information in more appealing way. Areas displayed are proportional to their relative frequency.



Two Ordinal Attributes

- Any of the methods previously described for bivariate analysis can also be used in the presence of two ordinal attributes.
- **Spearman’s rank** correlation should be used instead of the Pearson correlation.
- **Scatter plots with ordinal attributes** usually have the **problem** that there are many values falling at the same point, making it impossible to evaluate the number of values per point.
- In order to avoid this problem, some software packages **use a jitter effect** which **add a random deviation to the values**, making it possible to evaluate how large the cloud is.
- **Contingency tables** can be used and **mosaic plots** too. The values should be in increasing order.

Activity

Thank You